ASWATHY JAYAKRISHNAN

ASSISTANT PROFESSOR | AI/ML RESEARCHER

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PROFESSIONAL SUMMARY

Assistant Professor in the Artificial Intelligence and Data Science (AI&DS) Department at Adi Shankara Institute of Engineering and Technology, with an M.Tech in AI & ML and expertise in Artificial Intelligence, Machine Learning, and Deep Learning. Experienced in teaching Computer Networks and Networks Laboratory, mentoring students, and guiding academic and technical projects. Research interests include deep learning, computer vision, explainable AI, and AI-driven healthcare applications, with a strong interest in pursuing doctoral research.

TEACHING EXPERIENCE

Assistant Professor ***June 2026 - Present***

Adi Shankara Institute of Engineering and Technology,

- Teaching Computer Networks and conducting Computer Networks Laboratory sessions for undergraduate students.
- Delivering theoretical and practical instruction through lectures, laboratory exercises, demonstrations, and hands-on programming activities.
- Guiding and mentoring students in academic and technical projects, including problem identification, implementation, documentation, and presentation.

Teaching Assistant (Part-time) ***December 2025 – March 2026***

Rajagiri School of Engineering & Technology

- Assisted faculty members in conducting undergraduate laboratory sessions.
- Guided students in Python programming and Database Management Systems (DBMS).

TECHNICAL EXPERIENCE

Software Developer - Mastermine Technologies **July 2025 - June 2026**

- Developed a full-stack desktop application for a cross-platform application.
- Contributed to a multi-agent LLM analytics assistant for automated insight generation and decision support.
- Worked across Java / Spring Boot, React.js, and Electron.js and database integration in a production software environment.

Internship - ICT Academy Kerala ***May 2023 – June 2023***

- Developed a Twitter sentiment analysis system using Natural Language Processing techniques.

PUBLICATIONS

Aswathy Jayakrishnan, Lakshmi K S, Varghese S Chooralil, and Liyan Grace Shaji,
"An Evidential Hybrid Dual-Stream Framework for Diabetic Retinopathy Grading Using Fundus Images," International Conference on Cognitive Informatics Engineering and Technology (ICCET 2026), Vol. 1, p. 111, OSIET, Chennai, 28 March 2026,
ISBN: 978-81-69206-98-3.

Amrutha Muralidharan Nair, Sharon Susan Jacob, Anusha P S, Aswathy Jayakrishnan, Ashvin Manoj
"HalluChain: Measuring and Controlling Cascade Hallucination Propagation in Multi-Agent LLM Pipelines,"
2026 2nd International Conference on Innovations in Mechanical, Robotics, Computing and Biomedical Engineering (ICIMRBE), ERNAKULAM, India, 2026, pp. 1-5, **DOI: 10.1109/ICIMRBE70745.2026.11709899**.

EDUCATION

M.Tech, Computer Science and Engineering (AI & ML) ***Aug 2024 - May 2026***
Rajagiri School of Engineering and Technology
CGPA: 9 / 10

B.Tech, Computer Science and Engineering (Artificial Intelligence) ***Oct 2020 - Jun 2024***
Adi Shankara Institute of Engineering and Technology
CGPA: 8.05 / 10

TEACHING AND RESEARCH INTERESTS

Teaching: AI · Machine Learning · Deep Learning · NLP · Data Structures & Algorithms · Generative AI
Research: Multimodal Learning · CNN · Transformers · Medical AI · Explainable AI · Computer Vision

HARD AND SOFT SKILLS

Technical: Python, Java, C/C++, PyTorch, TensorFlow, Git
Soft: Teaching & Mentoring, Research Communication, Problem Solving, Team Collaboration

PROJECTS

X-AI- JackNet Explainable Deep Learning for Jackfruit Maturity Classification
Computer Vision | Explainable AI ***Jan 2026 - Present***
- Developed an explainable CNN-Transformer framework for fine-grained jackfruit maturity classification using attention mechanism and Grad-CAM

Ensembled Deep Learning Model for Explainable Diabetic Retinopathy Detection
Medical Image Analysis | Explainable AI ***Jul 2025 - Jun 2026***
- Developing an automated Diabetic Retinopathy (DR) grading system by fine-tuning state-of-the-art ConvNeXt and Swin Transformer models and developing Uncertainty Aware Hybrid model.

Occlusion based Face Expression Recognition
Computer Vision | Facial Recognition ***Jan 2025 – Apr 2025***
- Developed a hybrid deep learning model (EfficientNetB4 + Vision Transformer) for accurate, occlusion-resistant facial expression recognition, deployed via a Flask web app for real-time and uploaded video analysis.

VideoQuest: Video Q&A and Summarization Chatbot
Generative AI | NLP ***Nov 2023 – May 2024***
- Built and deployed a web application using Flask, T5, and DistilBERT that allows users to get summaries and ask questions about any YouTube video by processing its transcript.

People counting and tracking system
Computer Vision | Object Detection ***May 2023 - Jun 2023***
- Developed a crowd management system using OpenCV and YOLOv3 to monitor occupancy and automatically trigger alerts when capacity is exceeded.

CERTIFICATIONS

- AI-based Predictive Analysis for Disease Diagnosis and Healthcare Monitoring — Faculty Development Program, RSET, 2026
- Introduction to Large Language Models (LLMs) — NPTEL, IIT Madras, 2025
- Google Data Analytics Professional Certification - Coursera (2024)
- Prompt Engineering - Infosys Springboard (2025)
- Introduction to Generative AI Learning Path - Coursera (2025)
- Gen AI for UI/UX Designers - Coursera (2025)